Practical – 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

Theory :

1. Is the schema normalized properly?

Yes, the database is divided into three tables: students, events, and registrations. Each table stores related information. This reduces duplicate data and makes it easier to manage records.

2. Are primary keys and foreign keys correctly defined?

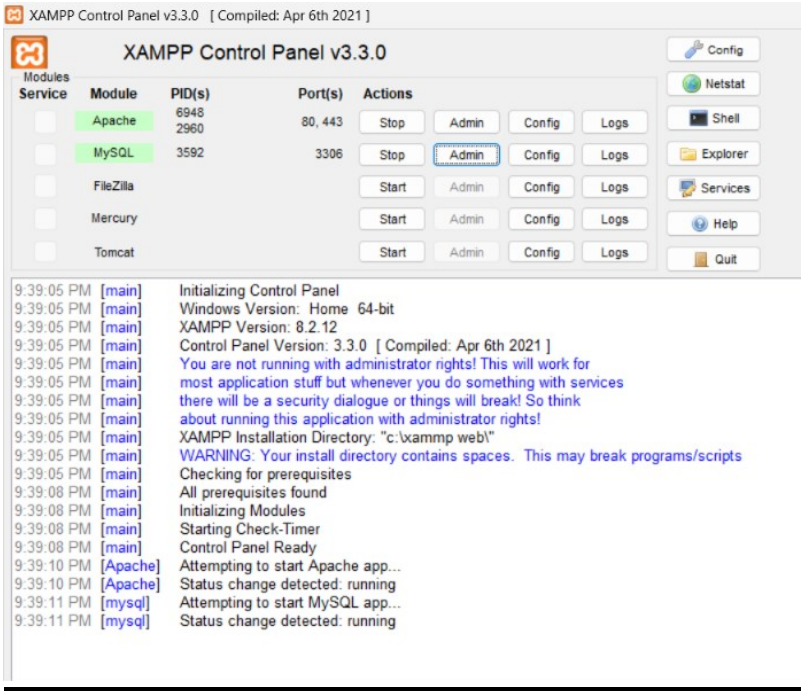
Yes, primary keys uniquely identify each record in a table. Foreign keys connect the students, events, and registrations tables. They help maintain relationships between tables and keep the data consistent.

3. Is the database connection handled securely?

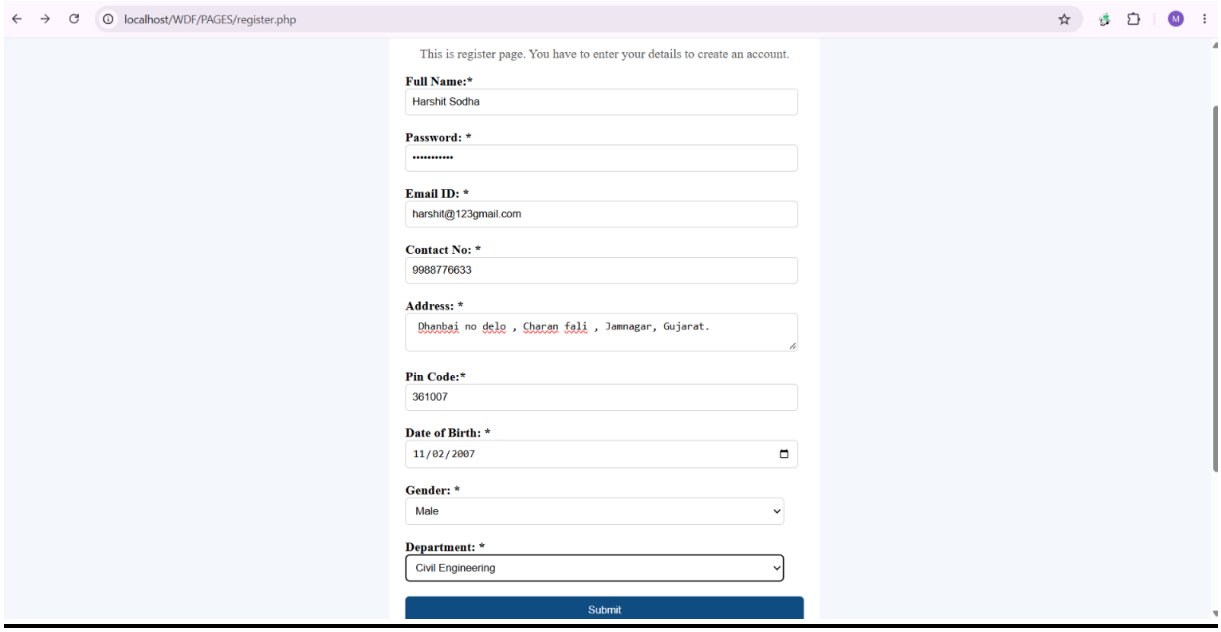
Yes, PDO is used to connect PHP with MySQL. Error handling is used to manage database connection problems. Prepared statements help protect queries from SQL injection.

Implementation/ Output/Result validation of password and Email id using Regex: (Pages with JS)

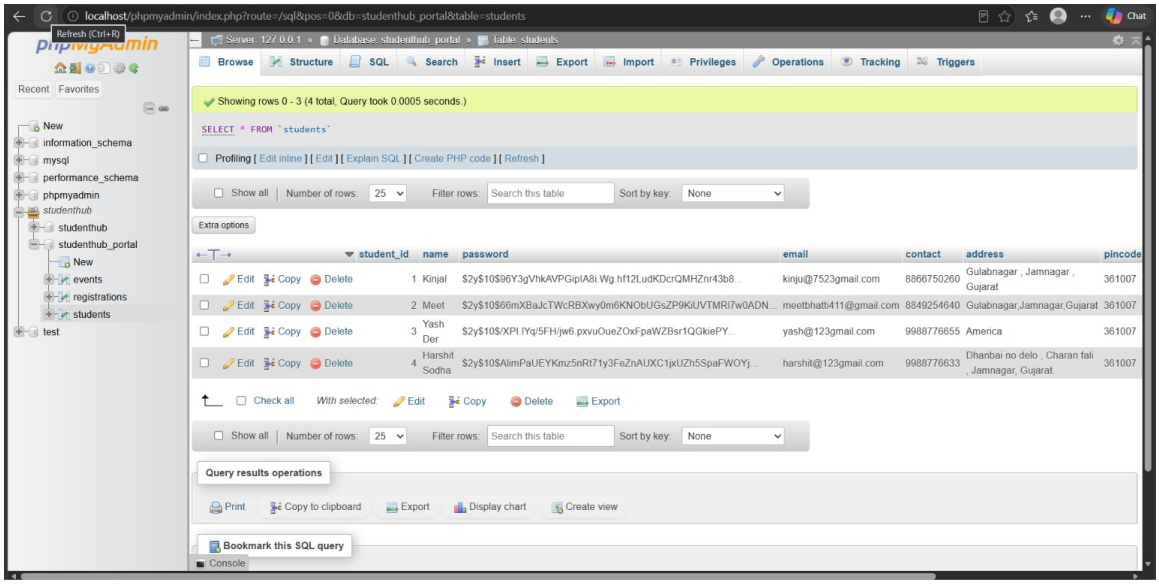
(1)Start XAMMP Local Host Server(APACHE and MYSQL Start):



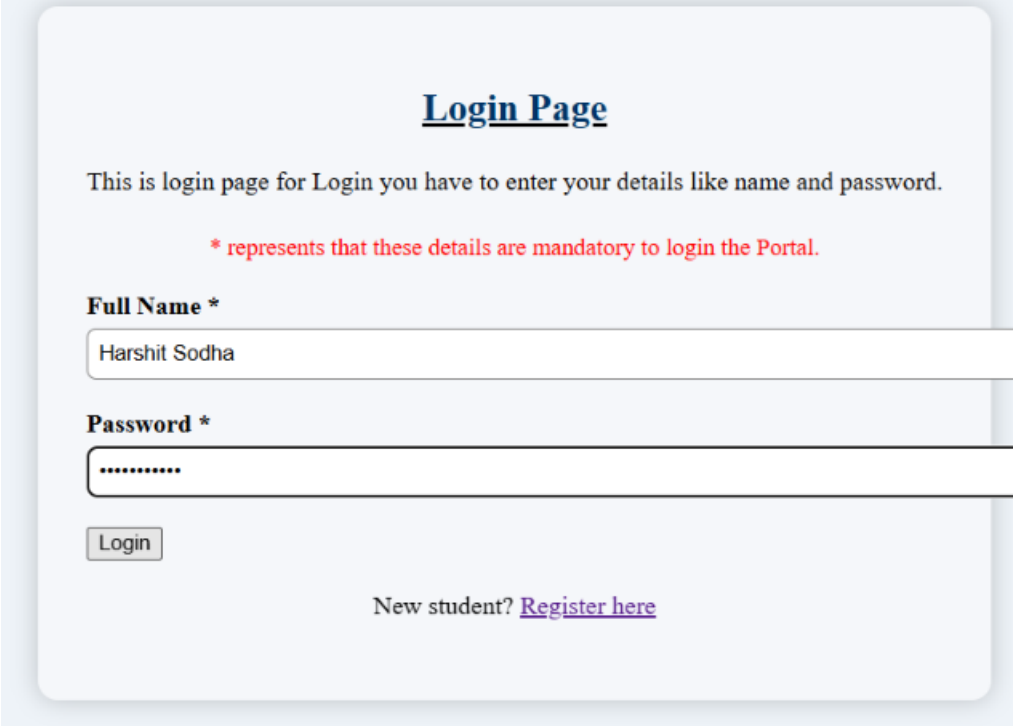
(2)Register with New Student:

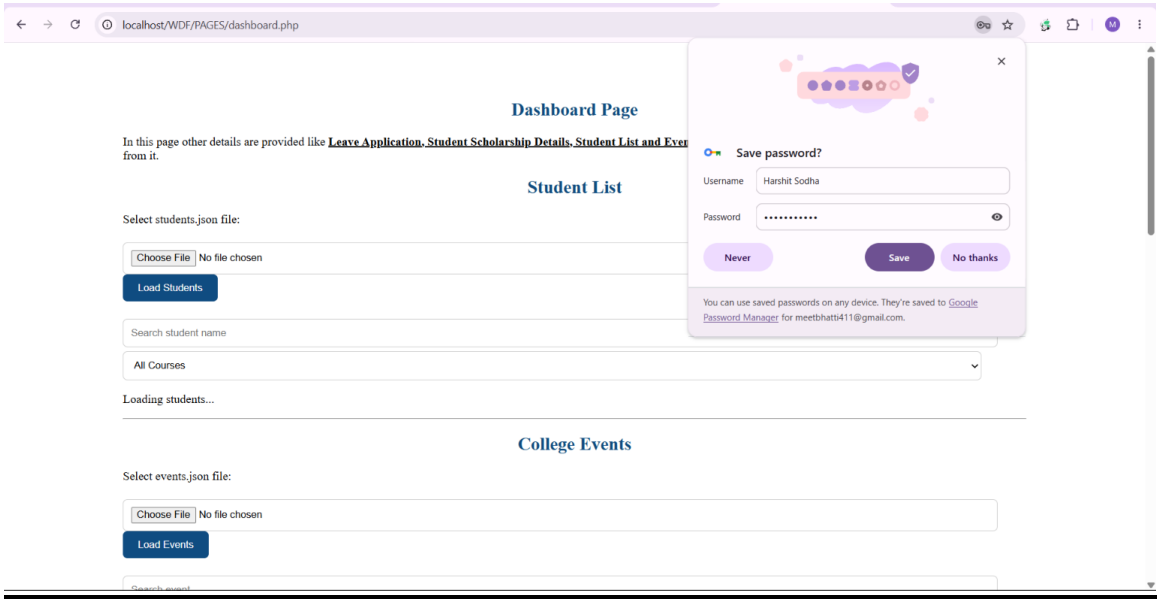


(3)After complete register see entry in database:

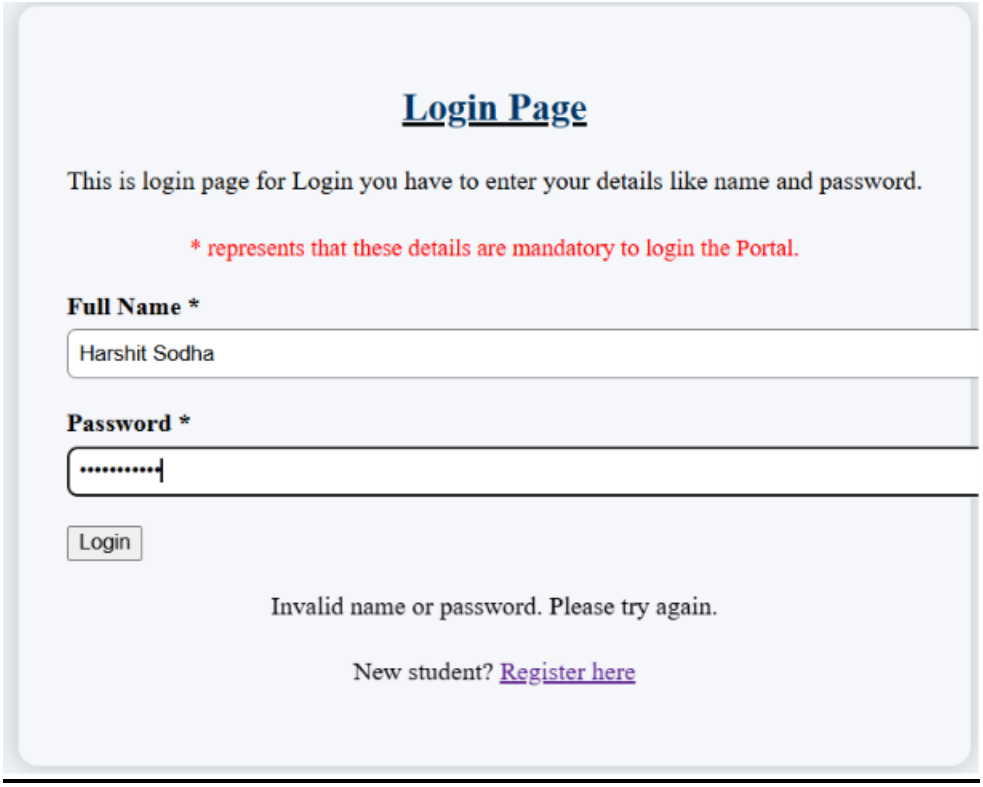


(4)Try to login with same username and password:

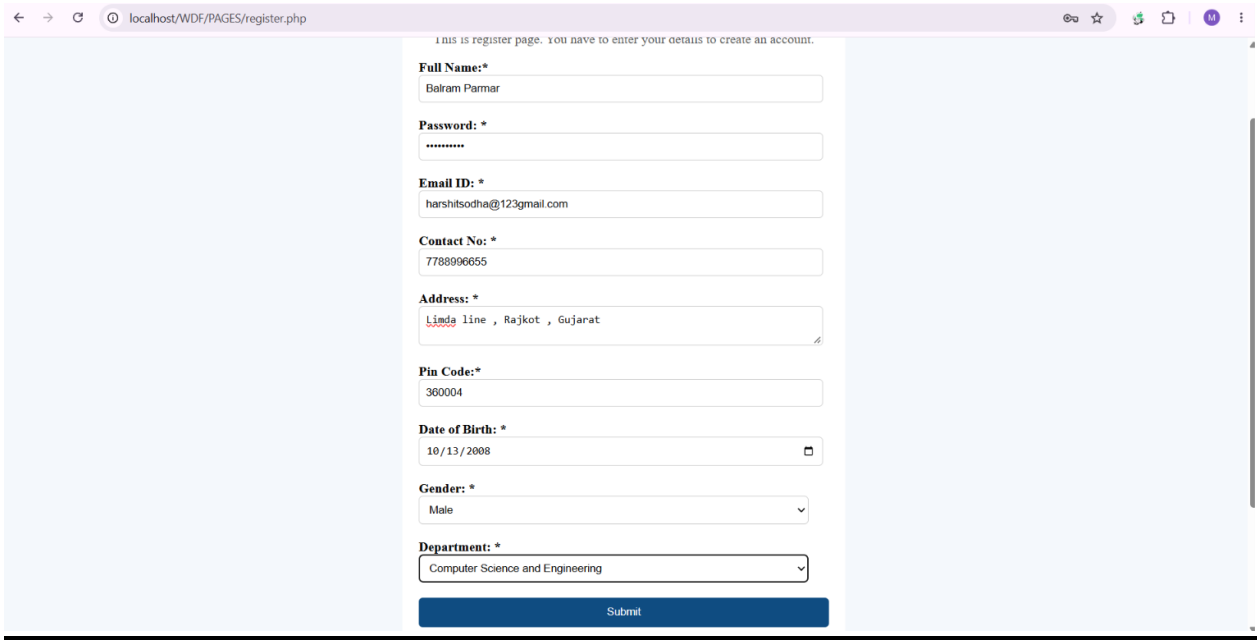




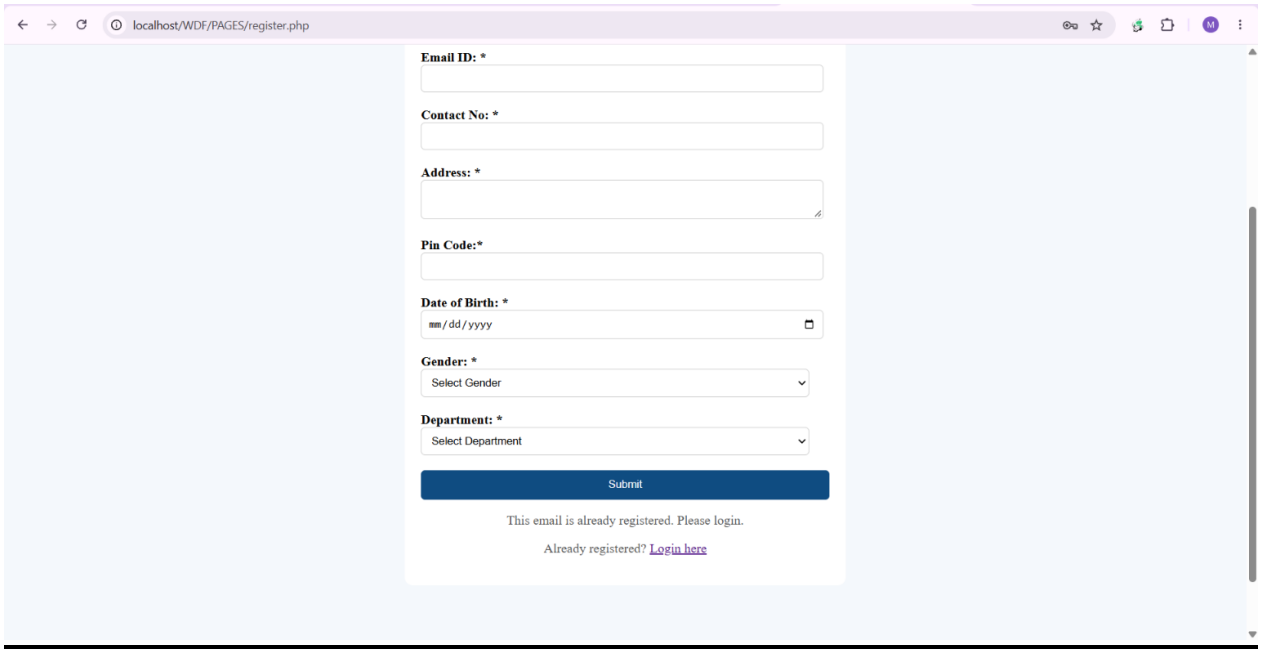
(5)Try to Login with wrong password:



(6)After login with correct username and password try to register new student with usable email id :



Error that Email is already registered:



GitHub Repository Link:

https://github.com/MeetBhatti/WDF.git

Learning Outcomes:

I learned how to design a database using MySQL and create relationships between tables. I also learned to connect PHP with MySQL using PDO and use prepared statements for safer database operations.

Faculty Remarks: (Keep this blank for faculty)